Shu Quan

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Research interests: reliable LLM agents · reasoning evaluation · alignment under domain constraints · world models & AI4Science.

Education

Peking University **Sept. 2024 – June 2027 (Expected)**
M.S. in Environmental Science, GPA: 3.61/4.0 Beijing, China

- **Research:** AI Safety & Safety Evaluation, PKU-Alignment Group (Advisor: Prof. Yaodong Yang)
- **Visiting Research:** LLM Hallucination Mitigation, HKUST(GZ) NLP-LAB (Advisor: Prof. Xuming Hu)

Henan University **Sept. 2020 – June 2024**
B.S. in Geographic Information Science, GPA: 3.68/4.0, Rank: 1/46 (Top 2%) Kaifeng, China

- **Relevant Courses:** Programming, Data Structures, Database Systems, Probability & Statistics
- **English:** IELTS 6.5, CET-6

Publications

* indicates equal contribution.

[1] Physics-Aware Video VAE via World-Model Alignment — AAAI 2027 (Under Review, Co-First, Lead)
Shu Quan*, [co-authors]

[2] Dense Clinical Contrasts Enhance Medical Knowledge Updating in Large Language Models — EMNLP 2026 (Under Review, Co-First)
[first author]*, Shu Quan*, [co-authors]

[3] A Blind Spot in Alignment: Quantifying Biosecurity Risks in Large Language Models — COLM 2026 (Under Review, Co-First, Lead)
Shu Quan*, Tianfang Hao*, Sitong Fang, He Geng, Jiayi Zhou, Boyuan Chen, Kaile Wang, Donghai Hong, Juntao Dai, Yaodong Yang, Jiaming Ji

[4] SafeCLI: Exploring System-Centric Risks in CLI Agents via Augmented Observability — NeurIPS 2026 D&B Track (Under Review)
Donghai Hong, Changhao Ju, Shu Quan, Yuyang Ren, Boyuan Chen, Jiayi Zhou, Kaile Wang, Juntao Dai, Yaodong Yang, Jiaming Ji

[5] SLM-MATRIX: A Multi-Agent Trajectory Reasoning and Verification Framework for Enhancing Language Models in Materials Data Extraction — npj Comp. Mater. 2025 (SCI Q1)
Xin Li*, Zhixuan Huang*, Shu Quan, Cheng Peng, Xiaoming Ma

[6] Nonlinear Effects of Blue-Green Space Variables on Urban Cold Islands in Zhengzhou Analyzed with Random Forest Regression — Front. Ecol. Evol. 2023 (SCI Q2)
Shu Quan*, Maojuan Li*, Tianqi Li, Haodong Liu, Yaohui Cui, Miaohan Liu

[7] Spatiotemporal Variations and Driving Factors of Atmospheric Environment in Central China — Atmosphere 2023, 14(1):92 (SCI)
Shu Quan, et al.

Research Experience

Physics-Aware Video VAE via World-Model Alignment **Mar. 2026 – Present**
Co-First Author (Lead) | Targeting AAAI 2027

- Diagnosed the physical-readability gap in modern video VAEs: latents reconstruct well but resist linear physical probing. Built a readout hierarchy (base / compact / grid+delta / rich) showing that the gap is structural, not capacity-bound.

- Designed a training framework that aligns the Wan VAE encoder to a frozen video world model (V-JEPA2) through a learnable projector and a hierarchical VF loss (token cosine, token–token geometry, temporal delta), with adaptive weighting to prevent representation collapse.
- Aligned latents improve Physion linear probe (0.7342 $\rightarrow$ 0.7485) and Physion++ OOD physical-property probe (+8% relative) while keeping PSNR drop $<$0.3 dB. Ablations reveal a non-monotonic relation between teacher-token similarity and physical readout, informing alignment-objective choice.

SEER-Bench: Dense Clinical Contrasts for Medical Knowledge Updating **Feb. 2026 – May 2026**

Co-First Author | EMNLP 2026 (Under Review)

- Reframed medical knowledge updating from "can the model update" to "how should the update be structured as supervision" under a same-budget SFT setting.
- Co-designed the EMQ (Extended Matching Question) supervision format, which exposes models to networked many-to-many decisions within a shared answer pool, in contrast to the isolated one-to-one structure of SAQ/FITB/MSQ.
- Co-built SEER-Bench, a temporally-anchored oncology-staging benchmark from the latest SEER Research Data release rendered against NCCN guideline updates; contributed input-density, representation-preservation, and cost-benefit diagnostics that explain EMQ's advantage. A Qwen3-4B trained on EMQ updates reaches 64.8% on temporally-anchored staging, outperforming four of seven external reference systems.

SPIKE-Bench: Quantifying Biosecurity Risks in LLMs **Jun. 2025 – Feb. 2026**

Co-First Author (Lead) | PKU-Alignment Group

- Built SPIKE-Bench, a benchmark and three-stage evaluation protocol (compliance filtering, biological plausibility via ESM-2/ESMFold, toxicity prediction) that measures whether LLM-generated amino acid sequences encode functional toxins — a risk invisible to natural-language safety evaluation.
- Audited 32 LLMs under this pipeline; Functional Harmfulness Rate (FHR) reached 50.7%, with near-zero correlation between refusal rate and functional risk ($\rho = -0.05$, $p = 0.79$).
- Trained BioSafe-Guard, a BioLinkBERT-based classifier achieving 98.9% detection with 0.3% false positives, reducing FHR across all 32 models to $\leq$0.5%.

SafeCLI: Exploring System-Centric Risks in CLI Agents via Observability **Oct. 2025 – Mar. 2026**

Co-author | PKU-Alignment Group

- Designed OS-Monitor, a kernel-level plug-and-play detection framework with 28 detectors across 7 system-risk categories, using auditd, procfs, and sysfs to produce step-level quantifiable safety signals; built the SafeCLI benchmark (314 Linux CLI tasks, 13 scenarios).
- Found that 74% of system-level harms produce no visible signal in terminal output; initial violations raise subsequent harmful action rate by 21.43% (broken windows effect) — revealing a systematic blind spot in trajectory-level CLI agent safety evaluation.

SLM-MATRIX: Multi-Agent Scientific Literature Extraction **Sept. 2024 – Sept. 2025**

Co-author | Published at **npj Computational Materials**

- Designed a multi-agent framework combining Mixture-of-Agents collaboration, generator–discriminator reasoning with MCTS, and dual cross-verification for structured data extraction from materials-science literature.
- Achieved 92.85% accuracy on BulkModulus and 77.68% on MatSynTriplet without model fine-tuning, outperforming single-path and conventional baselines.

Earlier Work — Environmental & Atmospheric Modeling **2022 – 2023**

First / Co-First Author | Henan University

- Nonlinear blue-green-space effects on urban cold islands (Front. Ecol. Evol., SCI Q2, co-first): random-forest analysis of multi-source remote-sensing data over Zhengzhou, identifying 140 m park width and 56% vegetation coverage as cooling-intensity thresholds for urban-planning design.

- Spatiotemporal variations of the atmospheric environment in central China (Atmosphere, SCI, first author): multi-source spatiotemporal modeling that built my foundational training in real-world scientific-data analysis — the same instinct that now drives my AI4Science and world-model work.

Research Internships

Alibaba Health | Research Intern | Skill Harness for Medical LLM Agents **Apr. 2026 – Jun. 2026**

- Worked on the skill-orchestration layer of a medical LLM agent — how the model decides which clinical tool/skill to invoke (guideline lookup, drug-interaction check, triage routing) and how to verify its output before returning it to the user.
- Studied failure modes of skill selection and grounding under realistic clinical queries; designed diagnostic signals that surface silent skill-misuse not visible in answer-level correctness — extending the "evaluation blind spot" theme of my research into the medical agentic stack.

OPPO AI Center, Large Model Algorithm Dept. | Research Intern **Jun. 2025 – Oct. 2025**

- Studied planning degradation and factual hallucination in long-form report generation; redesigned the outline-generation pipeline into a multi-step intent-decomposition, reflection, and verification chain. Ran controlled experiments across model backbones, sampling strategies, and reflection rounds to identify failure modes (missing subgoals, ill-posed retrieval queries, inconsistent structure). Planning satisfaction improved by 5%.
- Designed a factuality evaluation protocol covering numerical claims, citation accuracy, and contextual consistency; combined URL reachability, source-content matching, and code-execution checks to produce case-level supervision signals. Long-text factual accuracy improved from 60% to 78.5%.

iFlytek, South-China Research Institute | Research Intern **Feb. 2025 – May 2025**

- Designed the core algorithmic modules for fine-grained medical LLM alignment: a tripartite rubric schema decomposing response quality into three orthogonal dimensions: main proficiency S_1 (medical correctness, reasoning validity, completeness), excellence bonus S_2, and safety veto S_3 (hallucination, harmful content), with a lexicographic preference protocol enforcing S_3 as a hard constraint to prevent reward hacking.
- Developed the Explicit Criteria Injection paradigm: expanded preference pairs along rubric dimensions into multi-dimensional supervision signals to train a criteria-conditioned reward model; the RM steered GRPO training on a Qwen3-8B medical model, improving overall accuracy by 22.3% and safety compliance by 21.7% on expert-adjudicated benchmarks.

Honors & Awards

- **Kaggle Silver Medal**, LLM Adversarial Attack Competition (76/1692, Top 5%) 2025
- **Outstanding Graduate**, Henan Province 2024
- **National Scholarship**, China (Top 1%) 2023

Technical Skills

Programming: Python, PyTorch, Hugging Face Transformers, LangGraph, vLLM, Linux, Git

LLM Methods: Fine-tuning (SFT, LoRA), preference data construction, reward modeling (RLHF/GRPO), benchmark construction, LLM-as-Judge, multi-agent orchestration

Evaluation: Capability evaluation, adversarial evaluation, biosafety pipelines (ESM-2, ESMFold), agent trajectory analysis, representation probing (linear, k-NN, MLP)